



# Jérôme Eertmans

Research intern at KDDI Research, Inc. | Postdoctoral researcher and Teaching Assistant at UCLouvain.

📍 Belgium    ✉ jeertmans@icloud.com    🌐 www.eertmans.com  
📱 jérôme-eertmans    📞 jeertmans    🆔 0000-0002-5579-5360  
👤 Google Scholar

## Education

---

**UCLouvain**, Ph.D. in Electrical engineering Sept 2021 – July 2026  
Thesis title: Differentiable Ray Tracing for Radio Propagation Modeling. Available [here](#).

**UCLouvain**, M.Sc. in Electro-mechanical engineering Sept 2019 – June 2021  
Majored in mechatronics.

- I undertook a variety of course topics: telecommunications, open source software, numerical simulations, digital design of ICs, machine learning, mining pattern in data, etc., for a total of 112% workload.
- Graduated Magna Cum Laude.

## Experience

---

**Research Intern**, KDDI Research, Inc. – Fujimino, Saitama, Japan July 2026 – present  
Working on machine learning for indoor radio propagation modeling, within the radio propagation and devices group.

**Teaching Assistant**, UCLouvain – Louvain-la-Neuve, Belgium Sept 2021 – present

- Teaching duties (practical sessions, evaluation, ...): see list [here](#).
- Research on my Ph.D. subject under the supervision of Profs. Claude Oestges and Laurent Jacques.

**Visiting Researcher**, University of Bologna – Bologna, Italy Sept 2024 – Dec 2024  
Worked with Profs. Vittorio Degli-Esposti and Enrico Maria Vitucci on Machine Learning applied to Ray Tracing for Radio Propagation, as part of our 'Generative Path Sampling' research project. Led to two conference papers <sup>3,4</sup> and one journal paper<sup>1</sup>.

For additional experiences, see my [LinkedIn profile](#).

## Languages

---

**French:** Native

**English:** B2+/C1

**Dutch:** B1

**Italian:** A2

**Japanese:** Basics

## Projects

---

### DiffeRT

Python package with backend written in Rust.

- Built on top of [JAX](#) for high-performance and differentiable code
- Thoroughly tested and well documented
- Fully open access, along with its 2D variant [DiffeRT2D](#)

### Manim Slides

Presentation tool I developed to produce higher-quality presentations at conferences.

- Most of my presentations are created using this tool and can be viewed on [my personal website](#)
- Used by other researchers to disseminate their research, with examples available in the [documentation](#)

### More projects

You may find an extended list of my projects and contributions on my [GitHub profile](#) or on the [projects page](#) of my website.

## Publications

---

- [1] Transform-Invariant Generative Ray Path Sampling for Efficient Radio Propagation Modeling** Mar 2026  
**Jérôme Eertmans**, Enrico Maria Vitucci, Vittorio Degli-Esposti, Nicola Di Cicco, Laurent Jacques, Claude Oestges  
[arxiv.org/abs/2603.01655](https://arxiv.org/abs/2603.01655) (arXiv preprint, submitted to npj Wireless Technology)
- [2] Fast, Differentiable, GPU-Accelerated Ray Tracing for Multiple Diffraction and Reflection Paths** Oct 2025  
**Jérôme Eertmans**, Sophie Lequeu, Benoît Legat, Laurent Jacques, Claude Oestges  
[arxiv.org/abs/2410.14535](https://arxiv.org/abs/2410.14535) (arXiv preprint, accepted at EuCAP 2026)
- [3] Towards Generative Ray Path Sampling for Faster Point-to-Point Ray Tracing** Oct 2024  
**Jérôme Eertmans**, Nicola Di Cicco, Claude Oestges, Laurent Jacques, Enrico Maria Vitucci, Vittorio Degli-Esposti  
[10.1109/ICMLCN64995.2025.11140249](https://doi.org/10.1109/ICMLCN64995.2025.11140249) (2025 IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN))
- [4] Comparing Differentiable and Dynamic Ray Tracing - Introducing the Multi-path Lifetime Map** Oct 2024  
**Jérôme Eertmans**, Enrico Maria Vitucci, Vittorio Degli-Esposti, Laurent Jacques, Claude Oestges  
[10.23919/EuCAP63536.2025.10999736](https://doi.org/10.23919/EuCAP63536.2025.10999736) (2025 19th European Conference on Antennas and Propagation (EuCAP))
- [5] Fully Differentiable Ray Tracing via Discontinuity Smoothing for Radio Network Optimization** Mar 2024  
**Jérôme Eertmans**, Laurent Jacques, Claude Oestges  
[10.23919/EuCAP60739.2024.10501570](https://doi.org/10.23919/EuCAP60739.2024.10501570) (2024 18th European Conference on Antennas and Propagation (EuCAP))
- [6] DiffeRT2d - A Differentiable Ray Tracing Python Framework for Radio Propagation** June 2024  
**Jérôme Eertmans**, Claude Oestges, Laurent Jacques  
[10.21105/joss.06915](https://doi.org/10.21105/joss.06915) (Journal of Open Source Software)

- [7] **Min-Path-Tracing - A Diffraction Aware Alternative to Image Method in Ray Tracing** Mar 2023  
**Jérôme Eertmans**, Claude Oestges, Laurent Jacques  
[10.23919/EuCAP57121.2023.10132934](#) (2023 17th European Conference on Antennas and Propagation (EuCAP))
- [8] **Manim Slides - A Python package for presenting Manim content anywhere** Aug 2023  
**Jérôme Eertmans**  
[10.21105/jose.00206](#) (Journal of Open Source Education)

## Grants and Awards

---

- Research internship at KDDI Research, Inc. – Fujimino, Saitama, Japan** Sept 2024 – Dec 2024  
FRS FNRS *Accomplishment of a stay abroad.*
- Best Propagation Paper Award – Dublin, Ireland** Apr 2026  
EurAAP - EuCAP 2026.
- TICRA-EurAAP Travel Grant – Dublin, Ireland** Apr 2026  
EuCAP 2026 travel grant.
- Long-term Scientific Stay in Pr. Vittorio Degli-Esposti's team (FNRS) – Bologna, Italy** Sept 2024 – Dec 2024  
FRS FNRS *Accomplishment of a stay abroad.*
- Long-term Scientific Stay in Pr. Vittorio Degli-Esposti's team (WBI) – Bologna, Italy** Sept 2024 – Dec 2024  
WBI *Bourses d'Excellence WORLD.*
- STSM CA20120 - Visit to Pr. Vittorio Degli-Esposti – Cesena, Italy** Apr 2024  
Cost Action Short-term Scientific Missions.
- Participation in EuCAP 2023 – Florence, Italy** Mar 2023  
FRS FNRS *Participation in a scientific meeting.*

## Personal Achievements

---

- ADE Scheduler:** An open source alternative to UCLouvain's scheduling tool, that became the official university tool in Sep. 2024.
- Manim Slides:** An open source presentation tool, used by many for teaching, conference presentations, or thesis defenses.
- Scouts:** 6 years as a scout (Les Scouts) leader.

## Reviews

---

- EuCAP:** EuCAP 2026 (5 reviews)
- IEEE:** Access (5 reviews), Transactions on Vehicular Technology (7 reviews), Antennas and Wireless Propagation Letters (1 review), and Open Journal of the Communications Society (1 review)
- Journal of Open Source Software:** 1 review (see [#9384](#))
- Nature Partner Journals: Wireless Technology:** 1 review

## Referees

---

**Prof. Claude Oestges, UCLouvain:** [claud.oestges@uclouvain.be](mailto:claud.oestges@uclouvain.be)

**Prof. Laurent Jacques, UCLouvain:** [laurent.jacques@uclouvain.be](mailto:laurent.jacques@uclouvain.be)

**Prof. Vittorio Degli-Esposti, University of Bologna:** [v.degliesposti@unibo.it](mailto:v.degliesposti@unibo.it)